Compact arithmetic hyperbolic Coxeter polytopes: a dimension bound of 20 and a nine-field refinement

Coxeter Polytopes Frontier Project

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Abstract

We give a computer-assisted exclusion of compact arithmetic hyperbolic Coxeter polytopes in dimension21. Earlier internally completed field and dimension reductions leave only the actual ground field $\mathbb{Q}(\sqrt{5})$. The new ingredient is a normalized angle pool between distinct whole components of the same spherical type. Its total vertex cost is one nonnegative variable, while its local payments use actual component multiplicities. Complete contact-star coverage and previously proved D5/H4 incidence transfers permit an exact rational angle rule. Every allowed finite rank21 vertex forest has charge at most40. Independently evaluated native quadrilaterals have payment at least $60007431727/60000000000 > 1$; the inherited triangle payment is at least2. A strict h-vector inequality forces average vertex charge greater than40, a contradiction. With the earlier bound21 and the thirteen-field reduction, this gives the internally proved compact arithmetic bound20. Technical appendices give the complete compositions of further rank20 charge improvements, yielding dimension at most19 in nine specified ground fields. Their imported internal local geometric lemmas have explicit statements and a source-pinned supplement; the appendices are not a self-contained replacement for all their derivations. The argument is not fully formalized in Lean or externally refereed, and no compact example attaining either bound is asserted.

1 Statement and scope

A Coxeter polytope is a finite-sided convex hyperbolic polytope with facet dihedral angles integral submultiples of π. Arithmeticity refers to its reflection lattice and its actual Vinberg ground field. It is not a condition on a convenient overfield containing displayed Gram entries.

Theorem 1.1. *Every compact arithmetic hyperbolic Coxeter polytope has dimension at most20.*

Appendices A and B prove the scoped refinements Theorems A.1 and B.1: the dimension is at most19 when the actual ground field is one of

$$\mathbb{Q}(\sqrt{3}),\ \mathbb{Q}(\sqrt{6}),\ K_7,\ K_9,\ K_{11},\ K_{15},\ K_{16},\ K_{20},\ K_{24}, \qquad K_m = \mathbb{Q}(2\cos(2\pi/m)).$$

Here $K_{24} = \mathbb{Q}(\sqrt{2},\sqrt{3})$. This is a complete charge-composition proof with explicitly identified internal geometric inputs in its reproducibility supplement. The all-field theorem remains20; no ceiling19 is asserted here for $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{5})$ or $\mathbb{Q}(\sqrt{2},\sqrt{5})$.

The earlier eight-field ledger records the preserved eighteen-page predecessor. The current combined ledger, `manuscripts/supplements/compact_arithmetic_nine_field19_sources_20261020.json`, updates the parent-source pin and adds the cubic-seven appendix while preserving every earlier local proof source. The current reproduction entry point checks this combined ledger before and after building.

The proof has two inherited reductions and one new golden-field step. The completed internal bound21 excludes dimensions at least22, using dimension-specific rules through the published compact ceiling29. At dimension21 the complete field reduction leaves precisely $\mathbb{Q}(\sqrt{5})$. We exclude that last branch below. The full new composition inventory is `results/compact_arithmetic20_universal_composition_20261018.md`; the earlier exposition is `manuscripts/compact_arithmetic21.tex`.

Published inputs, internal geometric arguments, and finite exact checks are distinct. Published assumptions include compact acute simplicity, finite/affine and Lannér classification, the arithmetic cyclic-product criterion and field restrictions, simplest-type reflection arithmeticity, rational isotropy, and the nonnegative symmetric h-vector theorem [1, 2, 3]. The supporting-star reduction, whole-component normalization, incidence transfers and rational recurrence are internal proofs. Exact scripts certify explicitly finite obligations; a numerical optimizer only proposes a vector. No claim of external priority or complete proof-assistant verification is made.

The noncompact arithmetic bound21 is unchanged. Its dimension21 example does not establish compact sharpness. Neither the unrestricted compact bound29 nor any nonarithmetic or right-angled bound is altered here.

2 The strict counting threshold at rank21

A planar angle is an incidence (v, F) of a vertex and a two-face. In a simple n-polytope it corresponds to an unordered pair of incident facet normals, marked in the finite vertex diagram.

Lemma 2.1. *Suppose nonnegative planar-angle weights on a compact simple21-polytope satisfy* $\sum_{v\in F} \mathrm{wt}(v, F) \geq 5-k$ *on every* k*-gonal two-face. Then the total vertex charge is at least* $40f_0+550$*, so its average is strictly greater than40.*

Proof. Each vertex is on $\binom{21}{2} = 210$ two-faces. Double counting gives

$$\sum_v \mathrm{wt}(v) \geq 5f_2 - 210f_0.$$

For the symmetric nonnegative h-vector of the simplicial dual,

$$f_0 = \sum_{i=0}^{21} h_i, \qquad f_2 = \sum_{i=0}^{21} \binom{21-i}{2} h_i.$$

Pair i with $21-i$ for $0 \leq i \leq 10$. In $5f_2 - 210f_0$, their coefficient on $h_i = h_{21-i}$ is

$$5\left[\binom{i}{2} + \binom{21-i}{2}\right] - 420 = 80 + 5(i-10)(i-11) \geq 80.$$

The constant80 is their coefficient in $40f_0$. Every extra term is nonnegative, and $i = 0$ contributes $5 \cdot 10 \cdot 11\, h_0 = 550$ because $h_0 = 1$. Hence $5f_2 - 210f_0 \geq 40f_0 + 550$. □

It suffices to pay2 on triangles and1 on quadrilaterals; nonnegativity handles $k \geq 5$. An upper certificate equal to40 is enough for a contradiction. There is no need to manufacture a strict numerical margin.

3 Complete supporting-star coverage

We explain the geometric quantifiers needed for the new local weights. Use the Lorentz form of signature $(21, 1)$ and spacelike outward facet normals r_i with $(r_i, r_j) \leq 0$ for distinct walls. Chamber inequalities are $(x, r_i) \leq 0$.

Lemma 3.1. *Every positive-definite principal root set defines an actual face. A positive affine principal set is impossible in a compact chamber.*

Proof. Let B be the positive-definite principal Gram matrix and h a strict interior timelike vector. A positive-definite Z-matrix has entrywise nonnegative inverse, so $c = B^{-1}((h, r_i)) \leq 0$. The vector $h_0 = h - \sum_i c_i r_i$ is orthogonal to the chosen roots, has

$$h_0^2 = h^2 - ((h, r_i))^T B^{-1}((h, r_i)) < 0,$$

and has strictly negative product with every other facet normal. This gives a relative-interior point of the face. For an affine principal set, its positive kernel gives a nonzero null vector with nonpositive products against every wall. It cannot be zero because pairing its positive combination with h is negative. It gives an ideal point, contradicting compactness. □

Let W be the19 supporting roots of a quadrilateral and b_0, b_1, b_2, b_3 its cyclic side roots. Every adjacent extension of W is elliptic. Each opposite extension is indefinite by Lemma 3.1: otherwise the corresponding opposite sides would meet on this face. Minimal indefinite subdiagrams contain both opposite black roots and are Lannér pieces. Complete initial union supports retain shared white roots, black identifications and every cross contact.

Lemma 3.2. *A k-root contact core containing a positive corner of order $k - 2$ and a negative direction has nullity at most one. A complete quadrilateral star has nullity exactly one.*

Proof. The positive index is at least $k - 2$ and the negative index is one. For the full star $k = 23$ but the ambient Lorentz dimension is22, so it is singular and the nullity is exactly one. □

Proposition 3.3 (Final touching-star principle)**.** *For an unpaid actual quadrilateral, the inherited complete finite/dotted contact queues contain a final contact representative whose marked components are entire. Its remaining supporting components are common orthogonal spectators. Singular states that can extend must also remain stopping obligations.*

Proof. Begin with the actual opposite-pair minimal union. In each white component touching the entered core, order unentered vertices starting with a contact to that core and continuing along connected white edges. Every next root belongs to its complete one-white contact domain. Rejected arithmetic, signature or conjugate profiles cannot occur. A profile producing persistent basic payment pays the face independently of all adjustable bonuses. Otherwise a literal retained child map applies.

At a conditionally sealed state any further touching contact would give such persistent payment; hence the unpaid actual face has only orthogonal spectators. At an unsealed singular state continue while actual touching roots remain. Choose the final such state, rather than an earlier embedded singular subcore. Its contact part has corank one after the positive spectator block is split off, by Lemma 3.2. It has at most23 roots. The complete queues retain every singular stopping state, including expandable and boundary states, within their larger geometric root bound. Its marked components are therefore whole in the actual corners. The remaining components are orthogonal to the black and entered white roots. □

The finite contact domains and literal maps in this proposition are exact certificate premises with separate internal completeness proofs. The golden alphabet is $2, 3, 4, 5, 6, 10$. Dotted parameters are bounded using physical and conjugate positive-semidefinite Schur complements together with integral cyclic products, not by sampled distances. The finite queue has5172 states and51599 contact profiles; its2070 singular obligations include expandable states. The dotted queue has438 states and264 singular obligations. Their exact interfaces and all-size proofs are listed in the earlier proof map.

The rank21 native iterator combines the prior complete parameter-stress inventory with every fitting singular finite-queue state. There are1979 native faces. For an N-root contact row, common spectator rank is $23 - N$. Enumerate exact external counts (e, f, g, j) of $A_1, I_2(5), H_3, A_4$ with

$$e + 2f + 3g + 4j \leq 23 - N, \qquad 23 - N - e - 2f - 3g - 4j \neq 1.$$

There is no untracked rank-one type; A_2, A_3 realize every residual rank at least two. The same spectator forest is used at all four corners. This gives102917 cases. Exact component multiplicities determine all denominators. A numerical payment from an embedded old subcore is not substituted for the actual new angle.

4 The angle rule and its geometric exceptions

Write $q = (a, b, h, d)$ for the exact numbers of entire $A_1, I_2(5), H_3, A_4$ components. Old coefficients use the capped selector $c = (a, \min(b, 2), \min(h, 1), \min(d, 2))$. The fixed baseline on a whole connected component C is

$$B(C; i, j) = \mathbf{1}_{\mathrm{dist}_C(i,j) \leq 2} + \mathbf{1}_{C = H_4} \mathbf{1}_{\{\mathrm{dist}_C(i,j) = 3\} \text{ or } \{ij \text{ is the five-edge}\}}.$$

The complete76 minimal triangle markings have baseline at least2. Nonnegative connected-orbit bonuses are available on baseline-zero marked pairs. Rooted cross bonuses from $A_1, I_2(5), H_3$ use actual denominators $a, 2b, 3h$ and respectively exclude destinations among the earlier hub families. This makes these source families disjoint. The inherited rooted A_4 source bonuses have denominator $4d$ on nonhub destinations, and marked pairs in distinct A_4 factors have denominator $4(d-1)$ with their symmetric rooted-orbit numerator. Summing the latter over factor pairs gives $d/8$ times the16 ordered source-root-pair coefficients. Thus all old charges are additive over components at fixed context, including the A_4 pair rule.

4.1 D5 and H4 incidence transfers

These internal geometric lemmas are used with their complete written proofs and exact coefficient certificates.

Proposition 4.1 (Previously proved priority rules)**.** *Let W denote the old total charge and b the old extreme-E_8 pair bonus. There is a nonnegative angle rule preserving all required small-face payments with the following conservative allowances:*

Whole vertex forest	*Allowance*
$D_5 + 2E_8$	$W - \min(1, 2b)$
$A_1 + H_4 + 2E_8$	$W - t$
$A_1 + A_4 + 2E_8$, $A_5 + 2E_8$	$W + t$
$A_1 + B_4 + 2E_8$, $B_5 + 2E_8$	$W + t$
All other forests	W

Here $0 \le t \le 1$. The two D5 state inequalities must both be imposed.

Proof ingredients and scope. At an E8-triangle-active $D_5 + 2E_8$ vertex delete the D5 fork unit. Its fork pair never supports a triangle. The complete eleven endpoint patterns and the mixed-pattern exclusion prove that every affected quad retains a full ordinary unit, even under simultaneous deletions. In an inactive state delete the two extreme E8 bonuses instead. The complete consumer theorem retains all possible D5 spectator contacts and proves another basic unit survives, also under an active deletion at another corner. This gives respectively $W-1$ and $W-2b$, hence the maximum $W - \min(1, 2b)$.

For the H4 rule delete t from its extra five-edge unit at the target forest. Its full triangle domain includes all A1 contacts and handles high labels analytically. Exactly four potentially deficient patterns remain; restoring t on their actual source angles pays every triangle, including simultaneous H4 modifications. Every affected quad keeps an ordinary unit. The source angle capacities are one. For the apparent two-angle A5 source, two required neighboring H4 walls cannot coexist: their projections onto the full old positive vertex space leave a common one-dimensional future timelike complement and force scalar product $(3 - \sqrt{5})/3 > 0$, contrary to acuteness. The two whole E8 factors are retained throughout this argument. All longer faces remain nonnegative.

The full normal and field computations, endpoint completeness, consumer proofs and simultaneous payment arguments are in the linked D5/H4 packets; this proposition does not follow from counting abstract source orbits alone. These rules preserve persistent baseline pruning independently of the adjustable bonuses and compose with one another. Here persistence means that a pruned face still has a proved basic unit payment. It need not retain the identical old marked angle: the D5 fork angle is deleted, and the complete simultaneous consumer theorem supplies its alternative. □

4.2 The new same-hub pools

Lemma 4.2. *At a simple finite vertex with $k \ge 2$ whole copies of a rank-r type H, there are $r^2\binom{k}{2}$ planar angles whose marked roots lie in distinct copies. Assign each the nonnegative addition*

$$\frac{u_H(q)}{r^2\binom{k}{2}}.$$

The total new vertex charge is exactly $u_H(q)$. If $k < 2$ the family contributes zero.

Proof. Choose an unordered pair of components and one of the r roots from each. These are distinct unordered marked facet pairs and exhaust the family. Summing cancels the denominator. The full tuple q is a vertex invariant and is constant throughout this sum. □

We use $H = A_1, I_2(5), H_3$. Their per-angle denominators are $\binom{a}{2}$, $2b(b-1)$ and $9h(h-1)/2$. The three families are disjoint, and their nonnegative additions preserve paid triangles and long faces. Proposition 3.3 supplies actual whole components and actual counts when these pools pay an otherwise unpaid native quad. Its once-only new vertex cost is

$$\rho(q) = \mathbf{1}_{a\ge 2} u_{A_1}(q) + \mathbf{1}_{b\ge 2} u_{I_2(5)}(q) + \mathbf{1}_{h\ge 2} u_{H_3}(q).$$

All six special forests of Proposition 4.1 have $\rho = 0$.

5 Complete forest maximization and exact certificates

The full rank21 alphabet has68 irreducible types:

$$A_1,\dots,A_{21};\quad B_2,\dots,B_{21};\quad D_4,\dots,D_{21};$$
$$E_6,E_7,E_8,F_4,H_3,H_4,I_2(5),I_2(6),I_2(10).$$

Fix exact hub counts q and remove the four hub types from the residual alphabet. If $w_c(T)$ is the old additive cost of type T, define

$$U_c(0)=0,\qquad U_c(s)=\max_T\{U_c(s-\operatorname{rk}T)+w_c(T)\}$$

on reachable ranks; rank one is omitted. Every recurrence choice appends an allowed component, and every nonempty multiset has a final component to remove. Hence arbitrary repetitions are covered exactly. The ordinary budget adds fixed hub cost and $\rho(q)$ once to $U_c(21-a-2b-3h-4d)$, never once per residual component.

Two ordinary budget offers are replaced to exclude exactly the physical decreased forests. If a residual forest contains an off-target type, reserve one such component and fill the remaining rank by U_c. The pure H4/E8 rank20 possibilities are $5H_4$, $3H_4+E_8$ and H_4+2E_8; the first two remain unchanged and are imposed explicitly with A1. The only pure D5/E8 rank21 solution is D_5+2E_8, imposed in both states. All four positive sources are also evaluated with $+t$ even if they were not ordinary maximizers. No forest is discarded or assigned an independently favorable physical state.

Exact certificate 5.1. *The rational vector in* `results/fresh_compact21_self_hub_exact40_reconstruction_20261018.json` *has these exact properties:*

1. *All bonuses are nonnegative and* $t=14030552/21687803\in[0,1]$.
2. *All1979 native faces and102917 common spectator cases have quadrilateral payment at least* $60007431727/60000000000>1$.
3. *All68 allowed finite types and729 exact hub-count tuples, including the priority exceptions, have total vertex allowance at most40. The maximum40 is attained by* D_{21}.

The exact engine retains53495 distinct enlarged native inequalities. It appends new coefficients before deduplicating; projecting to old columns recovers51443 old classes. It reconstructs the510 old support rows, but full coverage uses the729 tuples and complete recurrence, not just those selected rows. There are243 capped groups and1659 auxiliary rank states. Numerical auxiliary values are discarded.

The single guarded IPM proposal had numerical objective 40.00000000000242. Rounding every positive value upward to the 10^{-10} grid gave the checked upper $40+3\cdot10^{-10}$. The solver-free final reconstruction changes precisely five coordinates to $0,0,0,36,39$ and retains every other rounded coefficient. The three zero residues were about 6–$8\cdot10^{-13}$; the two pool values were about $2.42\cdot10^{-12}$ above36 and39. These observations select a candidate only. Every final face and forest inequality is evaluated over the rationals, so no numerical tolerance is a proof premise.

Independent checker `scripts/check_compact21_self_hub_primal_independent_20261018.py` reconstructs the old orbit coefficients, new selectors, all pool counts, and the complete recurrence without importing the new model or producer. Fresh execution session59683/chunk7766fb verified the stated minimum payment and exact maximum40 over850 final forest offers. Its ledger is `results/fresh_compact21_self_hub_exact40_independent_check_20261018.json`. Source hashes identify the precise checked data; they do not establish the geometric completeness propositions.

Proposition 5.2. *No compact arithmetic hyperbolic Coxeter polytope with actual ground field* $\mathbb{Q}(\sqrt{5})$ *has dimension21.*

Proof. Apply the angle rule to each actual finite vertex forest. Proposition 3.3 and Certificate 5.1 pay every otherwise unpaid quadrilateral. The baseline triangle theorem and Proposition 4.1 pay every triangle. Every other two-face needs only nonnegativity. The exact forest certificate bounds each actual vertex by40. This contradicts Lemma 2.1. □

6 The thirteen fields and final composition

Let $K_m = \mathbb{Q}(\cos(2\pi/m))$. The dimension-at-least14 field restriction [1, 2] and the inherited complete field proofs give the following dimension21 branches. The shorter list valid only from22 is not used at21.

Actual field	Dimension21 exclusion
$\mathbb{Q}$	Simplest-type arithmeticity and Meyer isotropy
$\mathbb{Q}(\sqrt{2})$	Complete all-white bound20 proof
K_7	Complete cubic-seven bound20 proof
$\mathbb{Q}(\sqrt{3}), \mathbb{Q}(\sqrt{6}), \mathbb{Q}(\sqrt{2}, \sqrt{3})$	Reviewed high-label theorem
$K_9, K_{11}, K_{15}, K_{16}, K_{20}$	Same theorem, separately bound branches
$\mathbb{Q}(\sqrt{2}, \sqrt{5})$	Separate exact-field F25 proof
$\mathbb{Q}(\sqrt{5})$	Proposition 5.2

The inherited single-field binder retains all thirteen branches and all recursive certificate inputs. In particular its biquadratic F25 proof is not applied to the proper subfield $\mathbb{Q}(\sqrt{5})$. The rational field is excluded because the simplest-type rational indefinite form is isotropic in this dimension, so the arithmetic quotient cannot be compact.

Proof of Theorem 1.1. The earlier complete compact arithmetic bound21 excludes every dimension at least22. At dimension21 the thirteen-branch reduction leaves only the golden field. Proposition 5.2 excludes that field as well. Hence every compact arithmetic chamber has dimension at most20. □

7 Reproducibility and verification boundaries

The proof's finite objects are exact field/signature domains, literal contact maps, finite-diagram orbit coefficients, native face payments and forest maxima. Their explicit source and data pins are carried by the rational reconstruction certificate. Reproduction needs no optimizer:

```
python3 -B \
  scripts/fresh_compact21_self_hub_exact40_reconstruction_20261018.py \
  --verify
python3 -B \
  scripts/check_compact21_self_hub_primal_independent_20261018.py \
  --candidate \
  results/fresh_compact21_self_hub_exact40_reconstruction_20261018.json
```

Run these as separate sequential batches through `scripts/run_bounded_research_check_20261018.py`, whose live watchdog adds the59-second cap,1GiB process-tree RSS limit and disk/RAM gates. Each checker owns the shared exclusive lock. The first reconstructs the saved candidate and checks every enlarged face and forest; the second uses a separate implementation. Large inherited queue replays are documented in the bound21 proof map and must also be scheduled in bounded sequential batches. New final composition/hash checks bind their existing certificates; they do not freshly regenerate those classifications.

The remaining validation boundaries must be explicit. Published structural theorems may be assumptions in a formal interface. Internal whole-star coverage, incidence priority arguments and exact certificate logic must be formalized rather than silently imported as published axioms. A Lean theorem assuming all field bounds would certify only the last composition. The current result is an internally reviewed computer-assisted proof with rational scripts and an independent saved-primal evaluation; complete geometric Lean verification and external review are separate tasks. No new high-dimensional chamber or sharpness assertion follows.

A A rank 20 improvement in eight ground fields

Put $K_m = \mathbb{Q}(2\cos(2\pi/m))$ and let

$$\mathcal{K} = \{\mathbb{Q}(\sqrt{3}), \mathbb{Q}(\sqrt{6}), K_9, K_{11}, K_{15}, K_{16}, K_{20}, K_{24}\}, \qquad K_{24} = \mathbb{Q}(\sqrt{2}, \sqrt{3}).$$

Here and below a statement about K means equality with the actual ground field, rather than containment in a displayed overfield.

Theorem A.1. *A compact arithmetic hyperbolic Coxeter polytope whose ground field belongs to $\mathcal{K}$ has dimension at most 19.*

Theorem 1.1 already excludes dimensions at least 21. This appendix excludes dimension 20, without extrapolating its rank 20 calculations to other dimensions. It does not improve the all-field ceiling 20. In particular it makes no ceiling 19 assertion for $\mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{5}), \mathbb{Q}(\sqrt{2}, \sqrt{5})$ or K_7.

We give the complete composition of a new half-refund rule with the previous charge, its triangle and quadrilateral obligations, and the strict counting contradiction. The six local quadrilateral exclusions are imported *internal geometric theorems*, with precise statements below. Their complete proofs form the named reproducibility supplement; they are not published inputs or consequences of the numerical rank 21 certificate. Thus this appendix is a complete composition proof, not a self-contained replacement for every local geometric derivation. The machine-readable ledger is `manuscripts/supplements/compact_arithmetic_eight_field19_sources_final_20261019.json`. It records exact source bytes and roles, including the required spectator-E_8 coverage correction.

A.1 Inputs, normalization and the previous charge

All local roots in this appendix are unit outward normals. Distinct normals have nonpositive scalar product. A positive-definite root set defines an actual face, and a positive affine set is impossible, by Lemma 3.1. A compact acute chamber is simple. We use the finite and affine Coxeter classification, arithmetic cyclic-product integrality and strict positivity at nonphysical embeddings, and the nonnegative symmetric h-vector theorem as external structural inputs. Arithmeticity of reflection groups in the relevant dimensions is of the simplest type, as in the main field reduction.

One arithmetic observation will recur. Add an actual wall c to the *entire* positive rank 20 vertex diagram. At the old simple vertex the new wall inequality is strict, so the resulting 21 normals are

independent and their physical Gram matrix is Lorentzian. The projection cost q_c against the old vertex is therefore greater than 1 physically and less than 1 at every nonphysical embedding. If q_c is a rational expression in a cyclic scalar $z \in K$, then

$$K = \mathbb{Q}(z). \tag{1}$$

Indeed, if this were a proper subfield, another embedding of the totally real field K extending its physical embedding would repeat $q_c > 1$. If the cost is rational, the same argument forces $K = \mathbb{Q}$. This observation concerns the whole old vertex; it is not valid for an arbitrary embedded component with unexamined outside contacts.

The initial charge on a marked pair is one when its distance in its finite component is at most two, and zero otherwise. Add one more when the pair is an *entire* component $I_2(m)$ with $m \geq 6$. Call this the original charge. Its triangle payments are at least 2 and its quadrilateral payments at least 1 over $\mathcal{K}$, by the internal high-label theorem. Its complete finite-domain proof is an input, not a new assertion about an arbitrary arithmetic field.

We retain the previously proved whole-A, whole-D, and B reductions and their positive coupons. The inherited internal charge theorem used here has the following exact scope and consequences.

Proposition A.2 (Previous charge interface)**.** *There is a nonnegative charge on every actual rank 20 vertex over $\mathcal{K}$, obtained from the original charge by the reviewed whole-$A/D/B$ rules, with every triangle paying at least 2 and every quadrilateral at least 1. Its only vertex costs above 35 are*

$$D_r + D_{20-r} \quad (10 \leq r \leq 16), \qquad D_{12} + E_8, \quad D_{13} + E_7, \quad D_{14} + E_6,$$

each with cost 36. It leaves both fork/short-leaf units in these forests unchanged. Its other exact bounds are

entire vertex forest	*cost bound*
A_{20}, B_{20}, D_{20}	35
$D_{19} + A_1,\ D_{18} + I_2(m),\ m \geq 6$	35
$B_{19} + A_1$	139/4
at least three components	34.

The only positive coupon with at least three components is the $B_{18} + 2A_1$ quarter, whose complete cost is 133/4.

This is the composed prior theorem, not an application of a whole-D refund to a D factor with spectators. Its inputs are the whole-A two-end refund, whole-D three-unit refund, their mixed-consumer theorem, the B simple-end and balanced-quarter rules, the B_{20} far-four quarter, and the spectator rules for $B_{19} + A_1$, $D_{19} + A_1$ and $D_{18} + I_2(m \geq 6)$. All are identified in the ledger.

For clarity, the finite-forest part needs no new enumeration. A connected rank-r component has original cost $2r-2$ for D/E, for an entire high-label I_2, and for A_1 (cost zero); every other nontrivial component has cost $2r - 3$. Indeed a tree contributes its edge count plus $\sum_v \binom{\deg v}{2}$ distance-two pairs. Thus a forest with k components has original cost at most $40-2k$. For $k \geq 3$ the only positive coupon is the stated $133/4 < 34$ case. For $k = 2$ any nontrivial path component without a high-I_2 addition gives cost at most 35, apart from the separately bounded $B_{19} + A_1$. Two exceptional E components cannot fill rank 20. The only remaining cost 36 cases are $D + D$, $D + E$, $D_{19} + A_1$ and $D_{18} + I_2(m \geq 6)$; the last two have their own prior half-refunds. For $k = 1$ only A_{20}, B_{20}, D_{20} occur, already bounded by 35. This proves the stated frontier from the earlier consumer theorems.

A.2 The new rule and its complete support

At one of the ten cost 36 forests select the displayed larger D_r factor. At $D_{10} + D_{10}$ select either factor independently at each vertex. Write a for its fork, b, e for its short leaves and

$$a - l_1 - \cdots - l_L, \qquad N = r - 2 \geq 8, \quad L = N - 1,$$

for its long arm. Let T_s, $r + s = 20$, be the entire other D/E component. Reduce the two weights on (a, b) and (a, e) from 1 to 1/2. Leave the short-leaf pair (b, e) and every other angle unchanged.

The complete white support of a face carrying the pair (a, b) is

$$W = A_1(e) + A_L + T_s. \tag{2}$$

Its branched component T_s distinguishes it from all previous decreasing supports: the whole-A/D roles have A_{18} or $A_{17} + A_1$, the B roles have A_{18} or a white bond of label 4, $D_{19} + A_1$ has $A_{16} + 2A_1$, and the high-I_2 role has $A_{15} + A_1 + I_2(m \geq 6)$. The positive B coupons have white support B_{18}. None equals (2). Consequently no previous decrease occurs at any other corner of this face, and an original positive corner still contributes at least one unless it is a new half-source.

Among new roles the unique branched white component fixes T_s, and the path rank fixes r. Thus simultaneous new reductions on this face belong to the same family. In the tie case the whole white D_{10} already occupies its entire factor, so its roots cannot secretly supply a black selected pair. Omitting a reduction because a different factor was selected at another vertex can only increase payment. No consistent global choice of the selected factor is needed.

A.3 Every triangle retains two units

Let a, b, c be the sides of an actual triangle with a new source at (a, b). Both other corners are finite. In particular

$$W + a = A_{r-1} + T_s, \qquad A_{r-1} = e - a - l_1 - \cdots - l_L.$$

We first examine contacts to the whole spectator. A finite diagram is a forest, so c touches at most one vertex in each connected component. If c touches only T_s, a nonzero $b.c$ and the spectator contact must both be simple: a finite extension retaining a D/E fork has only simple bonds. The whole old-plus-c Gram data are then rational, contrary to (1). If $b.c = 0$ that whole rank 21 diagram is positive definite, impossible at positive index 20. If c also touches the long path, the whole finite corner of rank 20 is connected and contains T_s's fork, so it must be D_{20}. The spectator is D_s and c joins its long endpoint to a path endpoint simply. A nonzero $b.c$ would make a second fork in the other finite corner. At $b.c = 0$ the whole old-plus-c cost is again rational and physically greater than 1, forcing $K = \mathbb{Q}$. Hence

$$c \perp T_s \tag{3}$$

in every possible triangle. Orthogonality has been proved after examining spectator contacts, not assumed at the outset.

Finite extension of the whole path A_{r-1}, of rank at least 9, now leaves five cases: no path contact; a simple or label 4 contact at either endpoint e, l_L; or a simple contact at either next-to-end vertex a, l_{L-1}. Two path contacts create a cycle, and the exceptional finite diagrams cannot extend this path at its rank. Put $b.c = -\beta$. We show that each triangle remains paid even if its new fork units were deleted completely.

Near endpoint e. If $c.e$ has label 3 or4 and $\beta > 0$, the finite path $e - c - b$ allows ordered labels $(3,3),(3,4),(3,5),(4,3)$. The marks (a,c) and (b,c) have respectively distance two and one. Neither is a new source, so they retain one unit each. At $\beta = 0$, label 3 gives the affine E_8 tree with arms $1,2,5$ at a, using $a, b, e, c, l_1, \ldots, l_5$. Label 4 gives the affine block on a, b, l_1, e, c with positive null coefficients 2 at a, e, 1 at b, l_1, and $\sqrt{2}$ at c. Both are forbidden.

Far endpoint l_L. At $\beta = 0$, a simple contact makes the whole rank 21 diagram $D_{r+1}+T_s$ positive definite. A label 4 contact gives a positive null vector with coefficients 1 on a and the long arm, $1/2$ on b, e, and $1/\sqrt{2}$ on c. At $\beta > 0$, finite classification of the opposite path forces the far contact simple and $\beta = 1/2$ or $\sqrt{2}/2$. The first makes an affine simple cycle. For the second, the full old-plus-c projection cost is

$$q_c = r\beta^2/2 + \beta + 1/2.$$

The actual cycle puts $\sqrt{2}$ in K. Extend its sign-changing embedding to K; it gives $q_c' = r/4 - \sqrt{2}/2 + 1/2 > 1$, contradicting conjugate positivity.

Far next-to-end l_{L-1}. The contact is simple. At $\beta = 0$ there is an affine D vector: coefficients 1 on $a, l_1, \ldots, l_{L-1}$ and $1/2$ on b, e, l_L, c. At $\beta > 0$ the other finite corner would have fork arms $1, 2, L-2$ and rank $r - 1 \geq 9$, which is not a finite diagram.

Near next-to-end a. Here $a.c = -1/2$ and $c \perp W$. At $\beta = 0$ the star with center a and leaves b, e, l_1, c is affine D_4. Otherwise the two corners with pairs $(a,b),(a,c)$ are new fork roles, and the third is $W + I_2(m)$, where $\beta = \cos(\pi/m)$. The actual black cycle makes 2β integral in K, and

$$q_c = \frac{r\beta^2 + 2(r-2)\beta + r - 2}{2}, \qquad q_c - 1 = \frac{(\beta+1)(r\beta + r - 4)}{2}.$$

Strict positivity gives, at each nonphysical embedding,

$$-1 < \beta_i < -1 + 4/r \leq -3/5, \tag{4}$$

and the whole-star argument gives $K = \mathbb{Q}(\beta)$. Every field in $\mathcal{K}$ has degree at most 5. The elementary totient condition $\varphi(2m) \leq 10$ gives exactly

$$m = 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 15.$$

The prime divisors of $2m$ must be among $2, 3, 5, 7, 11$ and multiplying their Euler factors gives this complete list. Label 3 is affine A_2; labels 4,5 give the excluded exact fields $\mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{5})$. For $m = 7, 8, 10, 11$ use conjugate $\cos(3\pi/m)$, for $m = 9, 12$ use $\cos(5\pi/m)$, and for $m = 15$ use $\cos(7\pi/m)$. These are genuine nonphysical conjugates and violate (4): all are positive except $\cos(5\pi/9) > -1/2$. Only $m = 6$ remains. Its third corner is an *entire* $I_2(6)$ and retains two units, paying the triangle even if both fork units were removed.

No path contact. Now $c \perp a, W$. At $\beta = 0$ the whole rank 21 set is positive definite. Otherwise the integral two-cycle $z = 4\beta^2 = 2 + 2\cos(2\pi/m)$ satisfies

$$q_c = rz/8, \qquad K = \mathbb{Q}(z), \qquad 0 < z_i < 8/r \leq 4/5.$$

The complete condition $\varphi(m) \leq 10$ gives

$$m = 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 14, 15, 16, 18, 20, 22, 24, 30.$$

Labels 3,4,6 give $\mathbb{Q}$; labels 5,8 give $\mathbb{Q}(\sqrt{5}), \mathbb{Q}(\sqrt{2})$. For all remaining labels except 12, the following nonphysical residues are coprime to m, not ± 1, and have $k/m < 1/3$:

m	7	9	10	11	14	15	16	18	20	22	24	30
k	2	2	3	2	3	2	3	5	3	3	5	7.

Their $z_i = 2 + 2\cos(2k\pi/m) > 1$ violate the window. This keeps label 20 explicitly; it does not assume β belongs to its square field. Only $m = 12$ remains, with exact field $\mathbb{Q}(\sqrt{3})$. The entire $I_2(12)$ third corner retains two units. Thus every triangle, including all simultaneous new reductions, pays at least 2.

A.4 Quadrilaterals: complete reduction to six local lemmas

Two new half-sources already pay one. If a quadrilateral has exactly one new half-source and another original positive corner, that other corner retains at least one by the white-support separation above. We must therefore exclude just a quadrilateral a, b, c, d with its source at (a, b) and three other *original* zero corners. The zeros imply

$$a.d = b.c = c.d = 0, \qquad d.e = d.l_1 = 0. \tag{5}$$

The latter equalities hold because a contact to either root, both simply adjacent to a, would give a distance-two unit at (a, d). Put $b.d = -\beta$ and $a.c = -x$. Actual acute Coxeter walls give $x, \beta = 0$, a native finite cosine, or a value at least 1. We do not infer dotted ambient products merely from polygon opposition.

Finite extension of the *entire* $A_{r-1} + T_s$ at $W + a + d$ and (5) give precisely the first-d inventory in Table 1. A near attachment is excluded by the zero at (a, d). Two contacts in one old component create a cycle. A path-plus-spectator connected finite diagram has rank 20 and a fork, so it must be D_{20}: the bridge is simple and uses a D long endpoint. Neither an E spectator nor a second fork nor a bond of label 4 in that connected component is finite. The old path has rank at least 9, so no exceptional internal attachment is omitted.

Row	Contacts of d to the complete white graph
I	None
F	Simple far endpoint l_L
B	Far endpoint l_L with label 4
T	Simple far next-to-end l_{L-1}
J	One simple contact to T_s, no path contact
Bridge	Simple l_L and a long endpoint of $T_s = D_s$

Table 1: The complete first-wall inventory for three original zero corners.

Here are the exact local geometric inputs. Each statement has the entire old forest and all hypotheses (5); it excludes every actual finite continuation $W + c + d$, not just an embedded classical subdiagram.

Proposition A.3 (Internal six-row exclusion)**.** *For $10 \le r \le 16$, $r + s = 20$, $T_s = D_s, E_6, E_7, E_8$ and $K \in \mathcal{K}$, none of the six rows in Table 1 can be an actual compact quadrilateral with three original zero corners. Both finite and nonfinite ambient opposite products are included. The J statement includes the two A_7-to-E_8 continuations at $N = 8$.*

This proposition is an imported internal theorem assembled from the local sources below. We give its exact proof interface and principal calculations to make its coverage assessable. The source ledger provides the full individual derivations; accepting a smaller-rank extension of an old D_{19}/D_{20} estimate is not one of the premises.

The common determinant identity. Project off the positive white block and write

$$\begin{pmatrix} s_0 & -1/2 & -X & -\rho \\ -1/2 & 1 & 0 & -\beta \\ -X & 0 & t & -\nu \\ -\rho & -\beta & -\nu & u \end{pmatrix}, \qquad s_0 = \frac{N+2}{4N}, \quad t, u > 0, \quad tu > \nu^2.$$

There are 22 actual roots in Lorentz dimension 21, with a whole positive corner of rank 20, so this matrix has determinant zero. When $\Delta = \beta^2 - u > 0$, eliminating the b, d block gives

$$X = \frac{\nu(\beta/2 + \rho)}{\Delta} + \sqrt{\left(t + \frac{\nu^2}{\Delta}\right)\left(s_0 + \frac{\rho\beta + \rho^2 + u/4}{\Delta}\right)}. \tag{6}$$

The other algebraic root is negative: the constant term of the quadratic has the sign forced by $s_0 > 1/4$. Thus the positive root is determined geometrically. For $\beta \geq 1$ its right side is nonincreasing: the relevant derivative numerators are

$$-\beta^2/2 - 2\rho\beta - u/2, \qquad -\rho\beta^2 - \rho u - 2(\rho^2 + u/4)\beta,$$

both negative. Replacing β by 1 is therefore an upper estimate.

Row I: actual dottedness and the complete contact split. Here $b, d \perp W$ and are also perpendicular to the projected c side. The actual compact polygon has a positive-length c edge; its two adjacent side lines b, d are perpendicular to that same line at distinct points. Their outward product is $-\cosh \ell$, so $\beta > 1$ is derived from actual geometry. Put $\delta = 4\beta^2 - 4 \in \mathcal{O}_K$. The identity becomes

$$\left(\frac{(x+\kappa)^2}{t} - s_0\right)\delta = 1, \qquad q_c = 1 - t + 2N(x+\kappa)^2. \tag{7}$$

The complete c split is: no long-path or spectator contact, with arbitrary finite e contact; a long-path contact with every allowable e contact; spectator only; spectator plus a simple e contact; or a path/spectator bridge. The no-spectator branches include both $N = 8$ exceptional E_8 attachments. Their full no-long, finite-long and dotted-long unit arguments are retained as separate internal lemmas in the supplement. No bounded algebraic-integer search is an input.

For a spectator contact the residual squares after its projection are

T_s	attachment	t_T
D_s	long endpoint (any leaf for D_4)	$1/2$
D_5, D_6, D_7	short leaf	$3/8, 1/4, 1/8$
E_6	length-two arm endpoint	$1/3$
E_7	length-three arm endpoint	$1/4$
E_8	none	–

This follows from finite D/E arm lengths and unit-Gram determinant ratios. A spectator-plus-e contact must be simple and leaves only $(t, N) = (1/4, 8 \ldots 14), (1/8, 13), (1/12, 12)$. With the

integral actual-cycle scalar $A = 2x$ and $V = 2A + 1$, equation (7) respectively produces

t	D_*	$D_*\delta$
1/4	$2NA(A+1) - 1$	$2N$
1/8	$8NA(A+1) + N - 2 \quad (N = 13)$	52
1/12	$72A(A+1) + 11 \quad (N = 12)$	24.

In every row D_* is coprime to its displayed divisor, hence is a unit. Either $x = 0$ contradicts $\delta > 0$ directly, or its physical absolute value exceeds 1; the full-star inequality gives absolute value greater than 1 at every other conjugate too. This contradicts its unit norm. For a path/spectator bridge the whole finite component is D_{18}, its residual is $1/(2N)$, and its two endpoint roles give an odd unit D_* satisfying $D_*\delta = 4$; the same conjugate norm contradiction applies. The spectator-only row is settled by the coupled integral-square norm argument. These are the five complete contact alternatives in the named I composition lemma.

Rows F and T at finite β. The integral cycle 2β belongs to K. The full old-star costs are

$$q_F = r\beta^2/2 + \beta + 1/2, \qquad q_T = r\beta^2/2 + 2\beta + 1.$$

They give $K = \mathbb{Q}(\beta)$ and the common nonphysical window $-1/2 < \beta_i < 1/4$. The complete finite list is the $\varphi(2m) \le 10$ list above. Labels 3,5 force excluded fields. For the other labels a forbidden nonphysical residue is

m	4	6	7	8	9	10	11	12	15
k	3	5	5	3	7	3	3	5	11.

Negative entries have angle greater than $2\pi/3$; the positive exceptions exceed $1/4$, including

$$\cos(5\pi/12) = (\sqrt{6} - \sqrt{2})/4 > 1/4.$$

Also $\beta = 0$ fails the physical cost $q_d > 1$.

Row B at finite β: the weighted-square exception. The actual integral scalar is $A = 2\sqrt{2}\beta$, not necessarily β. Its full cost is $rA^2/16 + A/2 + 1$, giving

$$K = \mathbb{Q}(A), \qquad -8/r < A_i < 0.$$

If $\mathbb{Q}(A^2)$ were proper, the sign automorphism $A \mapsto -A$ would give the forbidden conjugate $-A_0 \le -\sqrt{2}$. Thus $K = \mathbb{Q}(A^2) = \mathbb{Q}(\cos(2\pi/m))$ and the complete $\varphi(m) \le 10$ list applies, including $m = 20$. All labels except 12 fail the necessary nonphysical square bound $\sigma(A^2) < 64/r^2 \le 16/25$: for $m = 5, 8$ the values $3 - \sqrt{5} > 3/4$ and $4 - 2\sqrt{2} > 1$ suffice; for the other nonrational labels use the residues $k/m < 1/3$ listed above, giving $4 + 4\cos(2k\pi/m) > 2$. For $m = 12$, $A = 1 + \sqrt{3}$, so its conjugate window leaves only

$$r = 10, \quad T_s = D_{10}, \quad \beta = \cos(\pi/12), \quad K = \mathbb{Q}(\sqrt{3}).$$

This necessary profile is not discarded. In it the complete opposite corner is $A_1 + B_8 + D_{10}$: c has no path contact, or a simple near-end contact l_1 with e absent or simple. Eliminating b, d gives $R_0 = (2 + \sqrt{3})/8 < 1/2$. For no path contact,

$$(x + \gamma/2)^2 = R_0(t_T - \gamma^2), \qquad \gamma = -c.e.$$

The native gap leaves $x = 0, 1/2$. A spectator contact makes t_T rational, whereas the remaining possible equalities require $4 - 2\sqrt{3}$ or $1 - \sqrt{3}/2$ as residuals. Without a spectator, the last $x = 0$ case requires $4\gamma^2 = (20 + 8\sqrt{3})/13$, whose trace $40/13$ is not integral. For the near-end contact the exact identity is

$$(x + 3/8 + \gamma/2)^2 = R_0\left((12 + \sqrt{3})/24 - \gamma^2\right), \qquad \gamma = 0, 1/2.$$

A positive x makes the left side at least $49/64$, too large; at $x = 0$ the two right numerators are respectively $27 + 14\sqrt{3}$ and $15 + 8\sqrt{3}$ over 192, different from the left numerators 27 and 75. This closes the genuine finite exception.

Rows F, B and T for $\beta \geq 1$. For F the opposite path is A_N, for B it is B_N, and for T it is D_N. Their complete continuations are constrained by the whole finite corner, not by an assumed spectator seal. Equation (6) has the following exact data and bounds:

row	u	ρ	$Q(1)$ bound
F	$(N+1)/(2N)$	$1/(2N)$	$(N+3)/(2(N-1)) \leq 11/14$
B	$1/N$	$\sqrt{2}/(2N)$	$(5+\sqrt{2})/14 < 1/2$
T	$2/N$	$1/N$	$(N+6)/(4(N-2)) \leq 7/12,$

where $Q(\beta)$ is the second square-root factor. If c has no path contact, $(x + \gamma/2)^2/t \leq Q(1)$. The spectator table gives $t \leq 1/2$, or $t \leq 1/4$ with a simple e contact. Native gaps, the induced affine $D/E/B$ witnesses, and the full-star exact fields $\mathbb{Q}$ or $\mathbb{Q}(\sqrt{5})$ exclude these cases.

For B and T the only path contact is simple at l_1, with $\gamma = 0, 1/2$ and no spectator contact. In T it gives $X \leq 11/16$, while a positive x gives $X \geq 15/16$. In B the corresponding upper bound is $X < 1/8 + \sqrt{2/7} < 7/8 < 15/16$. Thus $x = 0$, giving affine D_5 at $\gamma = 0$ or the simple affine four-cycle at $\gamma = 1/2$.

For F the full path-contact list is a simple or label 4 near endpoint, simple near-next or far-next, or a simple near-end/spectator bridge. The respective estimates force $x = 0$ with an affine witness, or contradict even its zero-contact lower bound. For example the simple near-end case has

$$X \leq 9/112 + \sqrt{22}/7 < 15/16, \qquad X \geq 15/16 \quad \text{if } x > 0;$$

the near-four and bridge upper bounds are respectively $9\sqrt{2}/112 + \sqrt{11}/(7\sqrt{2}) < 7\sqrt{2}/16$ and $9/112 + \sqrt{11}/14 < 7/16$, below their lower bounds at $x = 0$. The sole shared-far-end alternative instead gives

$$x \leq 1/2 + \sqrt{(N+3)/(2(N-1))} < 3/2.$$

Its integral scalar $A = 2x$ generates K and has all other conjugates in $(-1/2, 1/4)$, since $q_c = NA^2/2 + A + 1/2$. Finite native x fails the complete cosine list. For $x \geq 1$ and degree at least 3, $0 < |\operatorname{Norm} A| < 3(1/2)^{d-1} < 1$. In degree two a positive other conjugate also has norm below 1; a negative one forces norm -1 and trace 2, hence $A = 1 + \sqrt{2}$ and the excluded exact field $\mathbb{Q}(\sqrt{2})$. Degree one forces $\mathbb{Q}$. This is the all-height far-end argument, not a height-box census.

Row J, with the small exceptional continuations retained. A simple spectator contact has rational residual $u \leq 1/2$. The full-star integral square $z = 4\beta^2$ generates K and has $0 < z_i < 8u/r \leq 2/5$. Integer norms and traces exclude every finite native β except

$$\beta = \cos(\pi/12), \quad K = \mathbb{Q}(\sqrt{3}), \quad T_s = D_s \text{ at its long endpoint}, \quad 10 \leq r \leq 14.$$

This exception is kept in the local theorem. If both c, d touch the spectator, the complete joint finite table is $D_4 \to E_6, D_5 \to E_7, D_6 \to E_8$, with projected cross $\nu = 1/4$ and no path contact. Their remaining constants satisfy $s_0 \le 7/24$, $u, t_T \le 1/2$; an e contact is absent or simple. For $\beta \ge 1$, equation (6) gives $X < 5/6$ when $\gamma = 0$, and $x < \sqrt{13}/8 < 1/2$ when $\gamma = 1/2$. The remaining native labels have rational or $\mathbb{Q}(\sqrt{5})$ full-star costs and are excluded. At the finite exception only the $D_6 \to E_8$ joint row remains; its exact comparison gives $X < 4/5$, leaving $x = 0, 1/2, \sqrt{2}/2$, each excluded by its full-star cost.

If $c \perp T_s$, then $\nu = 0$ and

$$X^2 = tR, \qquad R = s_0 + \frac{u}{4(\beta^2 - u)}.$$

For $\beta \ge 1$, $s_0 < R \le (N+1)/(2N) \le 9/16$; in the finite exception $R = s_0 + \sqrt{3}/6$, strictly between $(N+1)/(2N)$ and $5/8$. The complete classical path/end/e inventory is excluded by the native gaps, affine witnesses and rational full-star costs in the J source. There are also two *additional* continuations when $N = 8$: c can attach simply to l_3 or l_5, making E_8 from the white A_7. No e contact is then possible. They have

$$t = 1/16, \quad X = x + \kappa, \qquad \kappa = 5/16 \ (l_3), \quad 3/16 \ (l_5).$$

For $\beta \ge 1$, $X \le 3/16$ excludes l_3; l_5 forces $x = 0$ and the induced affine D tree with its two forks at a, l_5. In the finite exception $X < 1/5$, so again l_3 fails, while l_5 forces $x = 0$ and the impossible rational value $R = 9/16$. These are the mandatory E_8 addendum cases.

The bridge row. The whole opposite white factor is $D_{18} + A_1(e)$. Thus c either misses D_{18} and has any finite e bond, or lengthens its long endpoint l_1 simply and has e absent or simple. The integral $V = 2\beta$ has $K = \mathbb{Q}(V)$ and $-4/r < V_i < 0$. Finite β gives $0 < |\operatorname{Norm} V| < 2(4/r)^{d-1} < 1$ for $d \ge 2$; degree one forces $\mathbb{Q}$. For $\beta \ge 1$, the determinant data are $u = \rho = 1/(2N)$ and

$$Q(1) = \frac{N+4}{2(2N-1)} \le 2/5.$$

No D_{18} contact leaves the native labels 3,4, excluded by the usual affine $D/E/B$ witnesses. A simple l_1 contact gives $X \le 3/80 + \sqrt{17/75} < 43/80$, below the $15/16$ lower bound when $x > 0$. At $x = 0$ its absent/simple e bond gives affine D_5 or the simple affine four-cycle.

Every affine witness above is principal on actual outward roots. Its positive null sum is nonzero by strict interior pairing and is feasible against *all* other walls by acuteness, including any outside spectator walls. Additional contacts cannot remove the ideal ray. The complete local proofs supply all finite-corner inventories and the subsidiary native comparisons summarized here. With Proposition A.3, the one-half-source/three-zero quadrilateral is impossible. Every affected quadrilateral therefore retains payment at least 1.

A.5 Vertex bound, strict contradiction and verification

The new charge remains nonnegative. Its triangles pay 2 by Section A.3; its quadrilaterals pay 1 by the preceding complete consumer argument. Every target forest loses exactly one: its two distinct original units become two halves. No positive coupon occurs on a target, and no unproved whole-factor deletion is applied through spectators. Every other forest is unchanged. Proposition A.2 consequently gives total charge

$$C \le 35 f_0. \tag{8}$$

On a k-gon, the payment is at least $5-k$: for triangles and quadrilaterals this was proved, and for $k \geq 5$ it follows from nonnegativity. At a simple vertex in dimension 20 there are 190 two-faces, so

$$C \geq 5f_2 - 190f_0.$$

The nonnegative symmetric dual h-vector satisfies $f_0 = \sum h_i$ and $f_2 = \sum \binom{20-i}{2} h_i$. Pairing coefficients by $h_i = h_{20-i}$ gives the exact identity

$$5f_2 - 225f_0 = \frac{5}{2} \sum_{i=0}^{20} (i-10)^2 h_i \geq 500, \tag{9}$$

because $h_0 = h_{20} = 1$. Hence $C \geq 35f_0 + 500$, contradicting (8). This excludes dimension 20 in $\mathcal{K}$. Together with Theorem 1.1, it proves Theorem A.1.

The rank 20 step is written mathematics, not a new finite enumeration or optimizer certificate. The source `formal/CoxeterProofs/ArithmeticEightField19.lean` has compiled and passed a separate printed-axiom audit of seven final theorems. It checks the paired coefficients, endpoint surplus 500, quarter-scaled cost update, elementary triangle/quad logic, the charge contradiction and composition with the previous ceiling 20. Its printed axioms are only `propext` and `Quot.sound`; the coefficient identity is axiom-free. Actual geometry, local consumer classifications, complete forest coverage and the identification of actual h-vector/charge data remain explicit premises. This conditional kernel check is not an end-to-end formalization of the geometric supplement.

Named internal supplement. The complete main consumer source is `results/arithmetic_rank20_two_component_d_fork_half_refund_20261019.md`; the prior whole-forest theorem is `results/arithmetic_rank20_complete_fourteen_family_frontier_20261019.md`. For the local theorem Proposition A.3, the full proofs are the following sources, all under `results/`:

Row	Complete source
I	`arithmetic_rank20_isolated_d_spectator_contact_obstruction_20261019.md`
F finite; T	`arithmetic_rank20_simple_d_finite_beta_and_far_fork_closure_20261019.md`
B finite	`arithmetic_rank20_far4_finite_beta_obstruction_20261019.md`
F/B large	`arithmetic_rank20_far_simple_and_far4_large_beta_closure_20261019.md`
J	`arithmetic_rank20_spectator_only_quad_obstruction_20261019.md`
J correction	`arithmetic_rank20_spectator_only_e8_coverage_addendum_20261019.md`
Bridge	`arithmetic_rank20_bridge_quad_obstruction_20261019.md`

The I composition explicitly imports its no-long, finite-long and dotted-long unit proofs; their source pins and subsidiary arguments are included in the supplement ledger. Its whole-spectator finite continuation table is `arithmetic_rank20_fork_spectator_continuation_research_20261019.md`. The ledger also identifies every inherited prior charge theorem, the separate written reviews and the conditional kernel evidence.

The source ledger distinguishes those external structural inputs, inherited internal charge theorems, new local internal lemmas and the conditional kernel records. To make this appendix self-contained at the geometric level would require integrating all imported no-long, finite-long and

dotted-long norm arguments and the preceding $A/B/D$ consumer proofs, rather than replacing their source references by review counts. No external refereeing, worldwide priority, sharpness or new polytope example follows from the present theorem.

B The cubic-seven ground field

Theorem B.1. *A compact arithmetic hyperbolic Coxeter polytope with actual ground field* $K_7 = \mathbb{Q}(2\cos(2\pi/7))$ *has dimension at most19.*

Theorem 1.1 supplies the dimensions at least21 tail. We exclude dimension20 here; arithmetic inheritance by arbitrary faces is not assumed. Together with Theorem A.1, this gives nine specified fields with ceiling19. The all-field ceiling remains20; no ceiling19 is asserted for $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{5})$ or $\mathbb{Q}(\sqrt{2},\sqrt{5})$.

Imported local geometric theorems. The following seven internal interfaces concern actual compact arithmetic20-polytopes over K_7. Every component qualification means an *entire* vertex component. A marked pair is one actual planar angle. These are geometric consumer theorems, not published assumptions or deductions from the rank21 numerical certificate.

Input	Exact charge interface
A4	Start with the distance-at-most-two unit and an extra entire $I_2(7/14)$ unit, with no D_5 bonus. An actual exceptional quad receives two A_4-endpoint/isolate halves. At most one half is active per whole A_4 at a vertex; all polygons are paid.
A	Delete both end first/third units at A_{20} and $A_{19}+A_1$; fund the surviving actual far-bridge quads at second-node/isolate roles. Include all mixed A/D orders, including isolate swaps.
B	Delete the simple-end first/third unit at B_{20} and $B_{19}+A_1$; fund its two surviving actual models at the specified A_2/isolate or simple-end second-node/isolate roles.
B4	At B_{20} retain 1/4 at its simple-end adjacent pair and 3/4 at its unique4-bond pair. Add the endpoint/isolate and isolate-pair quarters below. Include all D_{20}-whisker mixed zeros.
D20	Delete the two fork/short-leaf units and short-leaf-pair unit at whole D_{20}, with all simultaneous consumers included.
Ds	At $D_{19}+A_1$ or $D_{18}+I_2(7/14)$ retain 1/2 at each fork/short-leaf mark; leave the short-leaf pair unchanged.
DE	In D_r+D_{20-r}, $10\le r\le 16$, and $D_{12}+E_8$, $D_{13}+E_7$, $D_{14}+E_6$, retain halves at the two fork/short-leaf marks of the larger indicated D factor.

At $D_{10}+D_{10}$ fix either factor once per vertex; the theorem works for either choice. The seven complete sources, hypotheses, acyclic imports and exact source pins are the named supplement in `results/compact_arithmetic_k7_dimension19_charge_composition_20261020.md`. In particular A4 imports the full sealed-core theorem, including all outside white components; DE retains its spectator-E_8 correction. This appendix proves their complete composition, without repeating the local Schur, cyclic-integrality and conjugate-positivity proofs.

One simultaneous face payment. Apply all seven rules together. Receiving coupons in A and B are activated by the complete actual contact model and counted once per marked angle. All remaining weights are nonnegative. Every triangle keeps two units at its entire $I_2(7/14)$ corner; none of our decreases marks that component. The only shared decreasing white supports are

$A_{17}+A_1$	A_{20} end versus D_{20} fork,
$A_{16}+2A_1$	$A_{19}+A_1$ end versus $D_{19}+A_1$ fork halves,
A_{18}	B_{20} far4 versus D_{20} short-leaf pair.

A proves both first interfaces, retaining the D_{19} halves and both isolate-swapped orders. B4's complete21-row proof supplies the third. The other white supports are, respectively, $B_{17} + A_1$, $B_{16} + 2A_1$, B_{18}, $A_{15} + A_1 + I_2(7/14)$, and $A_{r-3} + A_1 + T_s$ with a whole branched spectator $T_s = D_s, E_6, E_7, E_8$. They are distinct: the last has its unique whole fork, the preceding one its high bond, and the B graphs retain4. None has the whole A_3 required by an active A4 coupon. Thus each other affected quad has its standalone consumer proof; two retained halves already pay1. Positive coupons only help. A quad with no decrease retains its A4 baseline payment. Every quad therefore pays1 under the single combined rule, and nonnegativity handles polygons with at least five sides.

All vertex forests and receiving costs. An original rank-r component costs $2r - 2$ for D/E, A_1 or entire $I_2(7/14)$, and $2r - 3$ for the other nontrivial finite types. The A4 capacity raises its cost only from5 to at most $11/2 < 6$. Hence a rank20 forest with at least three components costs at most34 before the new receiving coupons. Their *only* whole forests are

whole forest	baseline	decrease	receiving	upper
$A_{19} + A_1$	35	2	1	34
$A_{18} + 2A_1$	33	0	2	35
$B_{19} + A_1$	35	1	3/4	139/4
$B_{18} + 2A_1$	33	0	5/4	137/4
$A_2 + B_{17} + A_1$	32	0	2	34
$A_2 + B_{16} + 2A_1$	30	0	4	34

Here the two A second nodes give two or four halves. B has just one simple-end second node: its half roles contribute 1/2 or1, and its simple-endpoint/isolate or unique isolate-pair quarter adds 1/4. The A_2 roles give two or four units. These counts allow every role to activate; no favorable incidence choice is needed. None of these forests contains an entire A_4, so the old active coupon adds nothing.

One-component rank20 forests are only A_{20}, B_{20}, D_{20}; their charges are $37-2$, $37-1-3/4-1/4$, $38 - 3$, all35. With two components, a cost above35 requires both deficits from twice rank to be2. The complete rank equation then leaves exactly $D_{19} + A_1$, $D_{18} + I_2(7/14)$, the seven $D + D$ pairs and the three $D + E$ pairs listed above. Each has original cost36 and loses1 by Ds or DE. Two E factors cannot fill rank20. The two-component receivers were already counted explicitly. Thus *every* rank20 vertex has charge at most35.

The strict contradiction and proof boundary. For total charge C, polygon payments, simplicity and the symmetric nonnegative dual h-vector give

$$35f_0 \geq C \geq 5f_2 - 190f_0 = 35f_0 + \frac{5}{2}\sum_{i=0}^{20}(i-10)^2 h_i \geq 35f_0 + 500,$$

since $h_0 = h_{20} = 1$. This is impossible and proves Theorem B.1 using the existing higher-dimensional tail. The finite-forest argument uses no new enumeration or optimizer. Actual geometric consumer coverage remains the explicitly imported written supplement, not a new full Lean theorem. No worldwide priority, external refereeing, sharpness or compact example is claimed.

The current combined source ledger is `manuscripts/supplements/compact_arithmetic_nine_field19_sources_20261020.json`. It preserves the earlier eight-field packet and pins this parent source, both appendices, the seven K7 inputs and their complete composition. The current single-command reproduction is `python3 -B scripts/verify_manuscript.py compact20 -mode`

`core -pdf`. The module `formal/CoxeterProofs/ArithmeticK7Field19.lean` separately checks all thirteen quarter-scaled family budgets and the final count logic. Its capped compilation and six-theorem axiom audit pass, with only `propext` and `Quot.sound`; actual forest coverage and geometric payments remain explicit premises. The inherited `CompactHVector` object retains its documented legacy cache provenance rather than being relabelled as a new capped build.

References

[1] E. B. Vinberg, *Absence of crystallographic groups of reflections in Lobachevskii spaces of large dimension*, Transactions of the Moscow Mathematical Society47 (1984),75–112. Original text: `https://www.mathnet.ru/php/getFT.phtml?jrnid=mmo&option_lang=rus&paperid=444&what=fullt`.

[2] V. V. Nikulin, *On ground fields of arithmetic hyperbolic reflection groups*, Introduction and Sections2,3, `https://arxiv.org/abs/0708.3991`.

[3] M. Belolipetsky, *Arithmetic hyperbolic reflection groups*, Lemma2.1 and Sections1,7, `https://arxiv.org/abs/1506.03111`.

[4] Exact internal sources, geometric reviews and immutable certificate interfaces, listed in `results/compact_arithmetic20_universal_composition_20261018.md`, `results/fresh_compact21_self_hub_exact40_reconstruction_20261018.md`, and `docs/compact_arithmetic21_proof_map.md`.